

frameroute: what was built, what it proved, and what happens next

A complete record of an anchor-delta video pipeline that was audited, corrected three times, stripped to what survived measurement, and rebuilt around the one claim that held.

STATUS · library production-ready · site deploy-ready · API unlaunched

EVIDENCE · 2 lectures, 20 grounded queries each TESTS · 27 passing

SECURITY · 7 defects closed and verified

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01

Executive summary

The claim the project was named after is dead. The claim that replaced it is measured, replicated, and worth money.

ADVE began as “Anchor-Delta Video Embedding” — embed a few anchor frames with CLIP and *reconstruct* the rest from YOLO object-track deltas, advertised as 96.67% fewer encoder calls. Its own benchmark disproved it: the detector used to skip the encoder cost roughly twenty times the encoder call it avoided. The trained reconstructor was then beaten on retrieval by spherical interpolation with zero parameters, and has been deleted.

What survived is smaller and true: **you can send 3–4× fewer frames to a vision model without hurting retrieval** — measured against ground truth neither index produced, replicated on two lectures — which cuts a video-VLM bill by roughly 80%. The selection algorithm itself showed **no reliable advantage over plain uniform sampling**, so the product is the saving and the measurement, not the sampler.

The three numbers

- **3–4× fewer model calls** (67–74% fewer) with retrieval *better* than full compute.
- **~80% cost cut** — about \$15 per hour of video on GPT-4o-class pricing.
- **1 hour of video indexed in 5–13 minutes** on a laptop GPU, ~3× faster than encoding every frame.

02

What the project was

The original architecture ran a YOLOv8 detector and ByteTrack on every frame, built a spatial graph of object positions, and fed the graph deltas to a trained GRU (De1taReconstructoꝛV3, later a UALW variant) that synthesised the CLIP embedding of frames it never encoded. A “safety gate” enforced a 0.88 cosine floor.

Four independent problems were found:

FINDINGS FROM THE ORIGINAL AUDIT

FINDING	CONSEQUENCE
YOLO cost ~120 ms/frame to avoid a ~5 ms CLIP call	Measured wall-clock gain was −5.3% , not the claimed 60%; memory rose 17.8%
The 0.88 safety floor compared the prediction to the <i>anchor</i> , not to truth	A reconstructor that echoes the anchor scores 1.0 forever — the gate could not fail
Every accuracy figure was cosine similarity to a true CLIP vector	Unrelated frames from one fixed camera sit at 0.85–0.95; the metric could not fail either
Keyframe feature propagation is prior art	Deep Feature Flow (2017), Eventful Transformers (2023), video codecs — claims 1–6 were not novel

Separately, the pipeline **crashed on every real video**: anchor frames emitted embeddings shaped (512,) and reconstructed frames (1, 512), and stacking the two in add_batch raised a numpy “inhomogeneous shape” error. The unit test missed it because it fed uniform random vectors.

03

What it is now

Two components, cleanly separated, with the reconstruction machinery deleted.

frameroute — the router

Cuts the cumulative-novelty curve into equal-area segments and spends one model call per segment: static stretches collapse to one call, busy stretches earn more. The change signal runs on a 160×90 downscale at **1–2 ms/frame** with **no detector and no encoder in the hot path** — the direct correction to the 120 ms mistake. Two guarantees, both about

the signal rather than about meaning: a hard trigger that always selects a large measured change, and a maximum gap that leaves no stretch uncovered.

The search stack

Indexes a video across three orthogonal signals and merges them onto one timeline: visual (CLIP + FAISS), on-screen text (EasyOCR), and speech (Whisper). On-screen text as a first-class, time-fused search signal is uncommon and decisive for lectures, screen recordings and meetings — the content CLIP alone reads worst.

Indexing is now CLIP-only

LeanIndexer routes, then encodes exactly the selected frames. **Every stored vector is a real model call**; nothing is synthesised, so “is this embedding trustworthy?” is never a question. As a side effect the shape bug cannot recur.

```
# the whole architecture in four lines
track    = analyze_video("lecture.mp4")           # cheap change signal, no model
picks    = FrameRouter().select(track, budget)    # which frames earn a call
vectors  = clip.encode(frames_at(picks))          # real embeddings only
index.add(vectors)                                # searchable
```

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The evidence

Scored by bench/grounded_eval.py against ground truth *neither index produced*: the lecture’s own slide text. Queries are mined automatically from recurring slide terms (emissions, logsumexp, optimization); the relevant frames for a term are the ones whose slides contain it. No model judges its own output.

GROUNDING RETRIEVAL — 20 QUERIES PER LECTURE, TOP-5, MATCHED CALL COUNTS

LECTURE · ARM	CALLS	HIT@5	PRECISION@5
cities_and_decarbonization — full compute	400	0.450	0.400
cities_and_decarbonization — frameroute	104	0.750	0.320
cities_and_decarbonization — uniform	104	0.600	0.240
deep_learning — full compute	400	0.450	0.450
deep_learning — frameroute	131	0.650	0.430
deep_learning — uniform	131	0.750	0.460

What holds

- **Skipping frames does not hurt retrieval.** At 67–74% fewer calls, hit@5 was *higher* than full compute on both lectures. Replicated on independent ground truth. This is what the ~80% cost cut rests on.
- **Latency.** 1 hour of video indexed in 5–13 minutes on an RTX 4050 (4.5–12× realtime), ~3× faster than encoding every frame; change signal 1–2 ms/frame.
- **Zero-parameter beats trained.** Spherical interpolation between anchors outperformed the trained reconstructor, which was then removed.

Why full compute scores *worse*: its top-5 clusters on five near-identical frames from one moment, while a sparse index spreads its top-5 across the timeline and touches more distinct labelled regions. Useful, and worth stating plainly rather than presenting as magic.

05

What failed – published deliberately

The router is not better than uniform sampling

At identical call counts it won one lecture by **+25%** and lost the other by **–13%**. Two clips, opposite directions. This was the *named innovation*; on current evidence the ten-line baseline a customer would write themselves performs comparably.

Three quality metrics were discarded

- **Lenient yes/no VLM check** — reported “median 100% answer parity”. It counted two *different* slides as a match. Withdrawn.
- **CLIP timestamp-recall** — penalises returning the correct slide a few seconds off; an artifact on static lecture footage. Withdrawn.
- **Strict free-form VLM judge** — scored 20–35%, but marked a frame **7 seconds** from the reference a miss. Too harsh to trust.

Only the OCR-grounded evaluation survived scrutiny, and it is the one quoted above.

Limits of the evidence

- Two lectures, 20 queries each — a real signal, **not a published benchmark**.
- Absolute hit rates are modest because CLIP reads text in images poorly. The penalty is identical for all arms, so the comparison is fair even though the absolute number is low.
- Continuous-motion footage (traffic, busy streets, fast sport) is where routing helps least — untested for the cost claim, and the wrong place to sell.
- **No customer has ever run this on their own video.** That is the missing number.

06

Engineering record

WORK COMPLETED

AREA	OUTCOME
Indexing blocker	Embedding shape mismatch fixed; real-video indexing restored (was 100% failure)
Auth regression	Health and registration endpoints unblocked; test suite restored 0/8 → 8/8
Dead code	~1,660 lines removed: reconstructor ×3, UALW, safety gate, enterprise pipeline, tracker, ego-motion, slide detector, 10 orphan modules
Config	30 fields → 14; dead reconstruction settings deleted
Region-aware novelty	Rescued 47 localized events on MOT17; +0.125 recall at tight budgets
Benchmark harness	Decode-once cache (was $O(\text{builds} \times \text{full-decode})$); <code>grounded_eval</code> , <code>latency</code> , <code>cost_parity</code> , <code>answer_grade</code>
Packaging	<code>pip install frameroute</code> , MIT licence, console entry point, numpy+OpenCV only
Security	7 defects closed — see next section
Tests	27 passing + <code>frameroute selftest</code>

07

Production readiness

Ready

- frameroute router, CLI and Python API
- Measurement harness — what the audit runs on
- CLIP-only indexing pipeline
- 27 tests green, pip-installable, MIT

Not ready

- HTTP server: never load-tested, no tenancy isolation (/v1/stats exposes every task to every caller), API keys stored in plaintext SQLite
- Hosted API is unlaunched
- Evidence base is two lectures

Security defects closed

#	DEFECT	FIX
1	Arbitrary file read — /v1/frame used video_id as a path; frame_idx=-1 streamed any file whole	Baseline + extension allowlist + commonpath containment
2	Upload path traversal — ../../evil.sh wrote outside the uploads directory	One sanitised name used for storage, task id and returned id
3	Sports endpoints interpolated video_id into filenames	Sanitised at both call sites
4	Built-in dev key always accepted	Refused when ADVE_ENV=production and no key set (fail closed)
5	Wildcard CORS	ADVE_CORS_ORIGINS, localhost by default
6	Rate-limiter table keyed on a caller-controlled header, unbounded	Swept and capped
7	/v1/chat called a retired model	Configurable, current default

Verified by compile, 27 tests, and positive/negative controls: a real uploaded video still resolves — including when the client sends the original spaced name — while ../../etc/passwd, auth.db, server.py and non-video extensions are all refused. **Caveat:** files uploaded before the change whose names contained spaces or unicode must be re-uploaded or renamed.

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Market and buyers

The router’s benefit depends on *pacing*, not on industry. It pays where change is bunched in time and ties where motion is continuous.

SEGMENT	WHY THEY NEED IT	FIT
E-learning / course platforms	Large lecture libraries; searchable slides + speech; this is the footage we tested	Proven
Meeting / call intelligence	Screen shares and talking heads; on-screen text is where CLIP fails	Strong
Video-RAG / "chat with video" apps	Cost is dominated by frames × VLM price — the largest dollar saving	Strong
Screen recording / dev tools	Static screens with bursts of change	Predicted
Mostly-idle CCTV (warehouse, retail after hours)	Long static stretches with rare events — the router's ideal case	Predicted, untested
Traffic, busy streets, fast sport	Continuous motion — uniform sampling is as good	Do not sell

Competition

The **search** half is crowded by well-funded incumbents — Azure Video Indexer is nearly feature-identical, alongside Twelve Labs, Google Vertex and AWS Rekognition. Do not fight there. The **frame-routing** half has no standalone product: teams write uniform sampling inline, in ten lines. That is the opening and the warning at once — your real competitor is not a company, it is the code a customer would otherwise write themselves.

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Business model

The strategy in one line
Give away the sampler. Sell the answer. Monetise the API. Lock in with the index.

Because the algorithm is not defensible, the wedge is the thing nobody else offers: **the Frame Budget Audit** — “send us one hour of your video and we will tell you, measured on your footage, how few frames you can send before retrieval breaks, and what that saves you.” It needs no moat, runs offline, and produces the only proof that closes a deal: a number on the customer’s own data.

RUNG	WHAT	PRICE	PURPOSE
01	Frame Budget Audit	Free	Proof on their data; starts the relationship
02	Open-source router	\$0	Adoption; removes the incentive to rebuild
03	Hosted API	~\$2–4 / video-hour	Revenue
04	Multimodal index	Usage	Retention — re-indexing an archive is expensive to leave

Rejected: licensing. It requires defensible IP, and there is none. **Selling the company is an outcome, not a plan** — pre-revenue with no IP is an acqui-hire.

09a

Positioning and messaging

The sentence

“Stop paying your vision model to look at the same frame twice.” Everything else is support for that line.

Sell the cost, never the search

Video *search* is owned by Azure Video Indexer, Twelve Labs, Google and AWS — a fight a small team loses. Video *cost* has no standalone product. Your competitor is not a company; it is the ten lines of uniform sampling a customer would otherwise write. Beat that by being easier and by knowing the answer they do not: how far they can actually push it.

Lead with the audit, not the algorithm

The algorithm is not defensible and did not beat the baseline. The *measurement* is the differentiated asset: nobody rigorously answers “how few frames can I send?” on a customer’s own footage. That is the wedge, it needs no moat, and it produces the only proof that closes a deal.

Disclose the weakness on purpose

Outreach and the landing page both state that our router did not beat uniform sampling. This is not modesty — it is the strongest available signal that the other numbers are real. Every vendor claims a magic algorithm; almost none publish what failed. Technical buyers, who are the buyers here, notice immediately.

OBJECTION HANDLING

THEY SAY	YOU SAY
“We already sample at 1 fps.”	That is the baseline I measure against. The question is whether 1 fps is already too much for your content — on the lectures tested, 1 frame every 4 seconds held retrieval just as well.
“Why not just use the open-source one?”	You should, if you have the engineering time. People pay so they do not have to run it, and for the searchable index on top.
“How do we know it won’t miss things?”	You do not have to take my word — the audit measures it on your footage against your own slide text, and reports the miss rate rather than hiding it.
“Is the algorithm novel?”	No. Change-aware sampling is prior art. The value is execution, honest measurement and integration — I would rather tell you that now than have you find out later.

Marketing and channels

No ad budget, no sales team, and a copyable algorithm. That rules out paid acquisition and enterprise sales motions, and rules in a **bottoms-up, evidence-led, developer motion**.

CHANNEL	WHY IT FITS	COST
Direct outreach (start here)	Five named targets, free work offered, reply in days. The only channel that produces a customer this month.	Time
Open-source launch	Give away the copyable part; adoption <i>is</i> the moat. Show HN / r/LocalLLaMA / LangChain & LlamaIndex communities.	\$0
The honest benchmark post	"We tried to beat uniform sampling and failed" is more shareable than another benchmark win, and it builds the trust the product is sold on.	\$0
Cost calculator	Self-serve "what is your video bill?" — a lead magnet that qualifies people before they talk to you.	\$0
Landing page	Converts the above into audit requests. Built and deploy-ready.	~\$10/yr domain
✗ Paid ads, conferences, outbound SDRs	Wrong stage. No proven conversion to spend against.	—

The funnel

Open-source repo and benchmark post → landing page → free Frame Budget Audit → hosted API → multimodal index (retention). Each step is free to the customer until the point where they are already saving money, which is the only honest place to start charging.

Content that is worth writing

- **"We measured whether skipping frames breaks video search."** The grounded-evaluation method, the result, and the failure. This is the credibility piece.
- **"Your video-AI bill is mostly duplicate frames."** The cost arithmetic, with the calculator attached.
- **"Three metrics we threw away."** Why the lenient VLM check, CLIP timestamp-recall and the strict judge were all wrong — genuinely useful to anyone evaluating video retrieval, and it demonstrates the rigour you are selling.

Timing

VLM prices are falling and your revenue is a share of what you save customers. The arbitrage narrows every month. This argues for speed and for capturing video-hours now, not for a better algorithm later.

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Money

Unit economics, from the measured call reduction at ~\$0.005 per VLM image:

CUSTOMER VOLUME	THEY SAVE / MONTH	YOU EARN / YEAR
1,000 video-hours / month	~\$15k	~\$44k
10,000 video-hours / month	~\$147k	~\$440k
50,000 video-hours / month	~\$735k	~\$2.2M

OUTCOME SCENARIOS — HONEST PROBABILITIES

SCENARIO	REQUIRES	PER YEAR	ODDS
No traction	Nobody pays	\$0	~50%
Lifestyle	1-3 small customers	\$50-150k	~25%
Real business	~10 mixed customers	\$0.5-2M	~15%
Scale	Become the cost layer	\$3-10M+	~5%

A sale is the likelier large outcome: **\$1-20M tuck-in** with proven tech and a few customers; **\$50-200M** only if you become the standard. Today, pre-revenue and without IP, the asset is worth very little on its own — **value is created by customers, not by more code.**

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Risks

RISK	SEVERITY	MITIGATION
No moat. Change-aware sampling is prior art and copyable in a week	High	Open-source it; compete on integration, evidence and the index
Falling VLM prices shrink the saving you take a cut of	High	Speed — capture video-hours while the arbitrage is wide

RISK	SEVERITY	MITIGATION
Customers build it themselves (it is ten lines)	High	Be easier than rolling their own; lead with the audit, not the sampler
Evidence is two clips	Medium	Replicate on customer footage — which is the sales motion anyway
Incumbents (Azure, Twelve Labs) already own video search	Medium	Do not sell search; sell cost
Server not production-grade	Medium	Audits run offline; do not launch the API until it is tested

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Roadmap

Immediate – not code

- **Push the repository.** 26 commits exist only on one machine; the stored GitHub credential is no longer accepted (password auth is retired). Create a Personal Access Token or switch the remote to SSH.
- **Rotate the four exposed API keys** (three Gemini, one MiniMax) — they were pasted in plain text.
- **Deploy the site.** `website/` is ready — drag it onto Netlify Drop. Then swap the contact address (four `mailto:` links) and replace the domain placeholders in `robots.txt` / `sitemap.xml`.
- **Send five emails.** Targets, template, follow-up, audit runbook and the deliverable format are all written in `G0-T0-MARKET.md`.

0–30 days

- Five audits delivered on real customer footage — the only missing number.
- Open-source the router publicly; publish the benchmark *including the negative result*.

30–90 days

- Convert two audits into paid hosted processing.
- Harden the server properly *only if* a customer needs the hosted path: load testing, tenancy isolation, hashed key storage.
- Replicate the grounded benchmark across five clips per domain.

Deferred by design

- Wiring the live server to LeanIndexer (internal; no prospect sees it yet).
- Budget-adaptive region gain (a TODO in `signals.py`) — worth doing only when tuning to a real customer's budget.
- A learned keep/skip router — only if parameter-free methods plateau on customer data.

The one metric

Video-hours routed through you per week. Nothing else predicts the outcome. If it grows, there is a business; if it does not after a quarter of trying, that is the answer — and it cost a quarter, not a year.

On method. Every figure here comes from a run that can be reproduced with the commands in the repository, on footage you supply. Three quality metrics were built and discarded before one survived; two published claims were withdrawn after better measurement contradicted them. That record is kept deliberately — a project whose numbers change when examined is worth less than one that examines them first.